Lecture 2: Inverse Trigonometric Functions

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

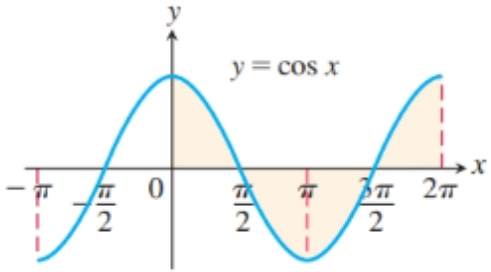
Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

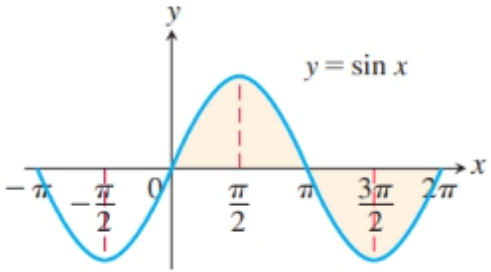
- ☐ Find and calculate the inverse trigonometric function.
- ☐ Find the derivative of the inverse trigonometric function.
- ☐ Perform basic integration using integration formula involving inverse trigonometric function.

Trigonometric Functions



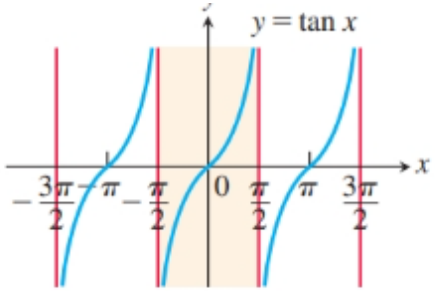
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(a)



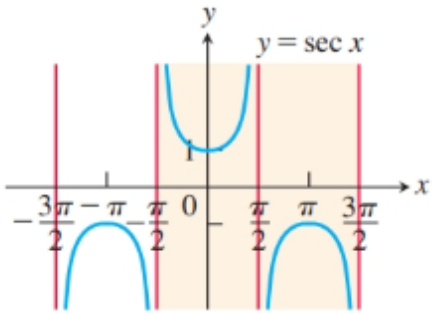
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(b)



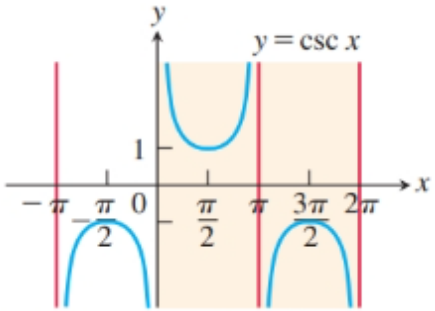
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $-\infty < y < \infty$
Period: π

(c)



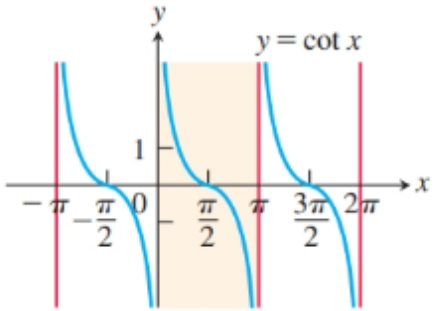
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $y \le -1$ or $y \ge 1$
Period: 2π

(d)



Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $y \le -1$ or $y \ge 1$
Period: 2π

(e)

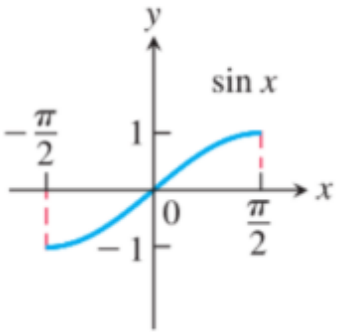


Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $-\infty < y < \infty$
Period: π

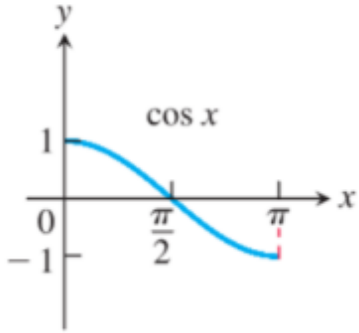
(f)

Generally, the trigonometric functions are not 1-1 function. However, as we learned in previous lecture, we can restrict their domain to make them 1-1 and hence have an inverse.

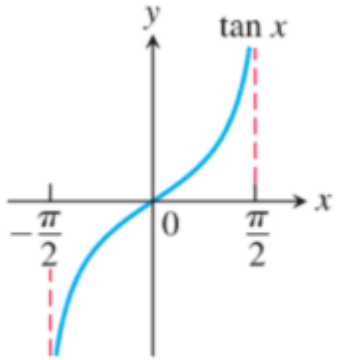
Trigonometric Functions with restricted domain



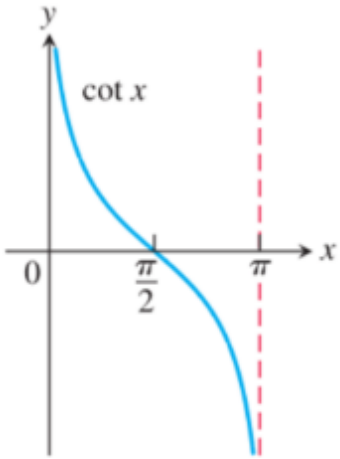
$y = \sin x$
Domain: $[-\pi/2, \pi/2]$
Range: $[-1, 1]$



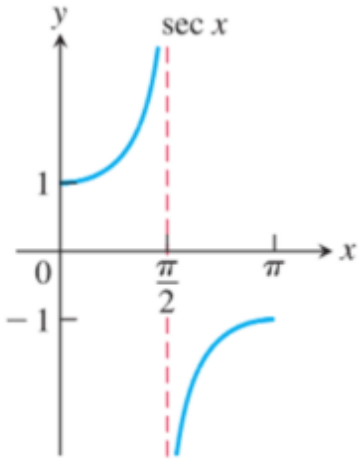
$y = \cos x$
Domain: $[0, \pi]$
Range: $[-1, 1]$



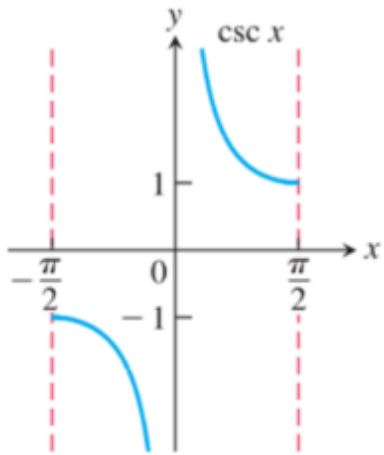
$y = \tan x$
Domain: $(-\pi/2, \pi/2)$
Range: $(-\infty, \infty)$



$y = \cot x$
Domain: $(0, \pi)$
Range: $(-\infty, \infty)$



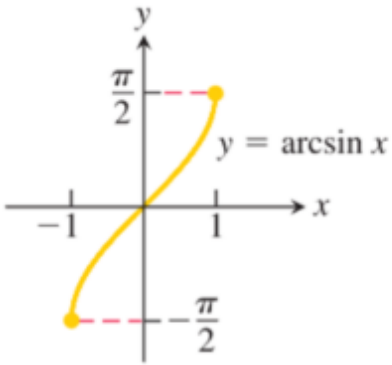
$y = \sec x$
Domain: $[0, \pi/2) \cup (\pi/2, \pi]$
Range: $(-\infty, -1] \cup [1, \infty)$



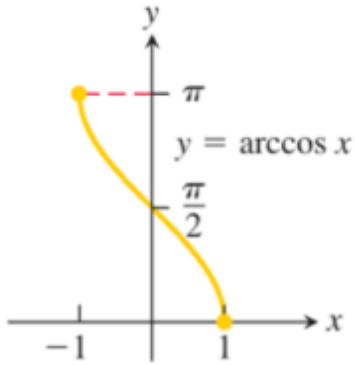
$y = \csc x$
Domain: $[-\pi/2, 0) \cup (0, \pi/2]$
Range: $(-\infty, -1] \cup [1, \infty)$

Inverse Trigonometric Functions

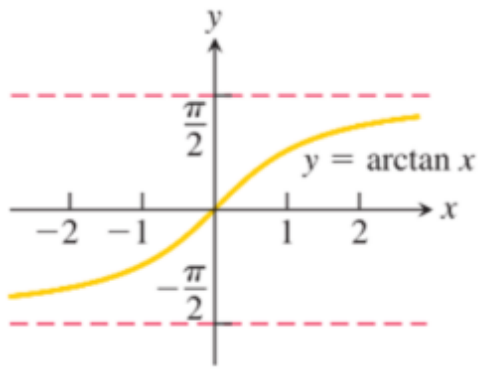
Domain: $-1 \le x \le 1$
Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$



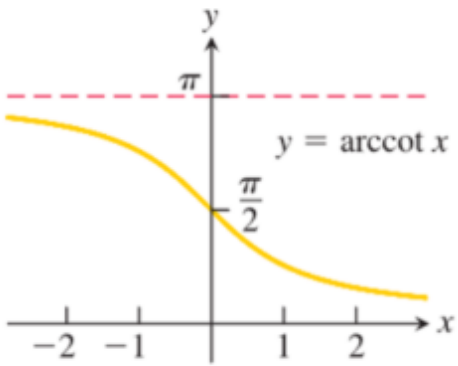
Domain: $-1 \le x \le 1$
Range: $0 \le y \le \pi$



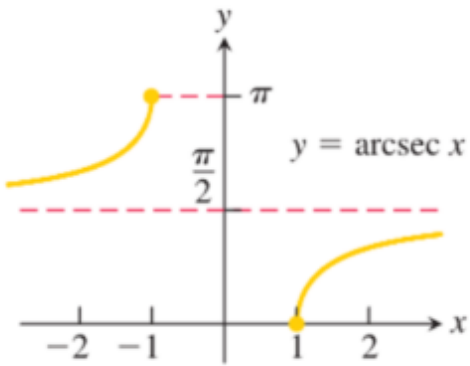
Domain: $-\infty < x < \infty$
Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$



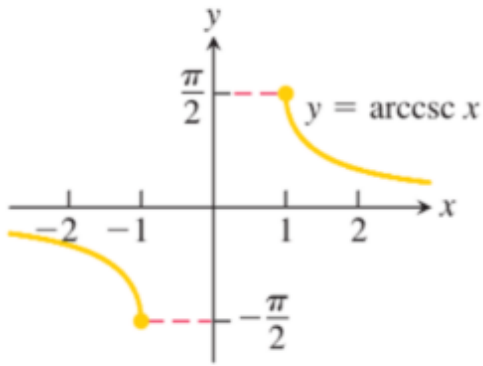
Domain: $-\infty < x < \infty$
Range: $0 < y < \pi$



Domain: $x \le -1$ or $x \ge 1$
Range: $0 \le y \le \pi, y \ne \frac{\pi}{2}$



Domain: $x \le -1$ or $x \ge 1$
Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}, y \ne 0$



Inverse Trigonometric Functions

These properties of inverse trigonometric functions can be useful

- $\arcsin(-x) = -\arcsin x$
- $\arccos(-x) = \pi - \arccos x$
- $\arctan(-x) = -\arctan x$
- $\arcsin(x) + \arccos(x) = \frac{\pi}{2}$

Inverse Trigonometric Functions

Example: Evaluate

- $\arcsin\left(\frac{\sqrt{3}}{2}\right)$
- $\arccos\left(-\frac{1}{2}\right)$
- $\arcsin\left(\sin\left(\frac{3\pi}{4}\right)\right)$
- $\cos\left(\arcsin\left(\frac{2}{7}\right)\right)$
- $\sin(2\arccos\left(\frac{3}{5}\right))$

Inverse Trigonometric Functions

Note that

$$\sin(\arcsin(x)) = \cos(\arccos(x)) = \tan(\arctan(x)) = x$$

is true for all x,

but

$$\arcsin(\sin(x)) = \arccos(\cos(x)) = \arctan(\tan(x)) = x$$

is only true for some x.

Derivative of Inverse Trigonometric Function

1. $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$ $(|x| < 1)$
2. $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$
3. $\frac{d}{dx}(\operatorname{arcsec} x) = \frac{1}{|x|\sqrt{x^2-1}}$ $(|x| > 1)$
4. $\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$ $(|x| < 1)$
5. $\frac{d}{dx}(\operatorname{arccot} x) = -\frac{1}{1+x^2}$
6. $\frac{d}{dx}(\operatorname{arccsc} x) = -\frac{1}{|x|\sqrt{x^2-1}}$ $(|x| > 1)$

Derivative of Inverse Trigonometric Function

Derivative of Inverse Trigonometric Function

Derivative of Inverse Trigonometric Function

Inverse Function-Inverse Cofunction Identities

$$\arccos x = \frac{\pi}{2} - \arcsin x$$

$$\operatorname{arccot} x = \frac{\pi}{2} - \arctan x$$

$$\operatorname{arccsc} x = \frac{\pi}{2} - \operatorname{arcsec} x$$

Derivative of Inverse Trigonometric Function

Derivative of Inverse Trigonometric Function

Example: Evaluate

1. $\frac{d}{dx}(\arcsin x^2)$
2. $\frac{d}{dx}(\frac{1}{\arctan x})$
3. $\frac{d}{dx}(\operatorname{arcsec}(5x^4))$
4. $\frac{d}{dx}(\arcsin(3x^4+5x+2))$
5. $\frac{d}{dx}(\ln(x)\arccos(x))$
6. $\frac{d}{dx}(\frac{2x}{\arcsin x})$

Integration Formulas

Integrals evaluated with inverse trigonometric functions

The following formulas hold for any constant $a > 0$.

1. $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin(\frac{x}{a}) + C$ (Valid for $x^2 < a^2$)

2. $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan(\frac{x}{a}) + C$ (Valid for all x)

3. $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\operatorname{arcsec}\left|\frac{x}{a}\right| + C$ (Valid for $|x| > a > 0$)

Integration Formulas

Example: Evaluate

1. $\int_{\sqrt{2}/2}^{\sqrt{3}/2} \frac{dx}{\sqrt{1-x^2}}$
2. $\int \frac{dx}{\sqrt{3-4x^2}}$
3. $\int \frac{dx}{\sqrt{e^{2x}-6}}$
4. $\int \frac{dx}{\sqrt{4x-x^2}}$
5. $\int \frac{dx}{4x^2+4x+2}$

Integration Formulas

Additional example: Evaluate

1. $\int \frac{dx}{\sqrt{4-9x^2}}$

2. $\int_0^1 \frac{dx}{x^2+4x+13}$

3. $\int \frac{4-x}{\sqrt{16-x^2}}dx$